

Project Report

Title of the Project

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Course Title: DEVELOPING ANDROID APPS

Submitted By:

Name: Devanjay Sinha

Reg No.: 12304899

Submitted To:

Dr. ANUJ SHARMA

UID: 20592



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DECLARATION

I, Devanjay Sinha, a student of Bachelor of Technology under CSE discipline at Lovely Professional University, Punjab, hereby declare that all the information furnished in this project report is based on my own work and is genuine

Your name: Devanjay Sinha

Registration Number: 12304899

ECOQUEST: GAMIFYING SUSTAINABILITY

Small Actions, Giant Impact — An Academic Project Report

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1. Executive Summary

Climate change and localized environmental degradation represent some of the most critical challenges facing contemporary society. While high-level awareness regarding sustainability concepts remains relatively elevated among the modern student demographic, a profound behavioral gap persists between theoretical understanding and active, everyday practice. EcoQuest is an innovative, production-grade mobile application engineered precisely to address this disconnect. By weaponizing gamification principles—including daily recurring challenge loops, verifiable crowdsourced proof structures, tier-based achievement mechanics, and localized institutional competition hierarchies—EcoQuest transitions sustainability from a passive ideology into an engaging, recurring habit loop.

This report provides a comprehensive academic analysis of the EcoQuest platform, outlining its thematic domain structures, architectural underpinnings, technological implementations, structural future roadmap, and initial validated environmental metrics. Built using a robust software engineering stack comprising Jetpack Compose for fluid native front-end rendering, alongside Google Firebase services for secure real-time state synchronization and scalable multi-role user access management, EcoQuest demonstrates how targeted software solutions can generate immediate, verifiable real-world impacts.

2. Project Description

2.1 Introduction and Vision

The primary objective of EcoQuest is to bridge the gap between environmental awareness and active daily execution. Traditional education models often fall short because they present sustainability concepts as abstract, macroscopic problems (e.g., global carbon emission projections) that leave the individual feeling detached or powerless. The underlying vision of EcoQuest flips this paradigm: it operates on the foundational principle that small, localized actions taken consistently by a collective community inevitably yield a massive, compounded macroscopic impact.

By targeting schools, districts, and university systems, EcoQuest cultivates a highly collaborative ecosystem. It fosters the next generation of 'Eco-Warriors' by translating complex environmental responsibilities into snackable, bite-sized daily challenges that can be successfully integrated into the frantic schedule of a modern student. Through digital prestige (badges, ranks, experience points) and social competition (localized class and school leaderboards), the platform ensures long-term behavioral retention and habitual growth.

2.2 Core Product Workflow & Architecture

The operational lifecycle of the EcoQuest app follows a strict, highly structured sequential loop designed to guarantee accountability, engagement, and reliable data logging. This loop consists of three primary, distinct architectural stages:

- **1. Action (Challenge Delivery):** The system provides a fresh, curated set of localized eco-tasks and challenges every single day. These tasks are designed to be bite-sized, categorized clearly by domain, and explicitly structured so they are achievable within home or school environments without interrupting standard student workflows.
- **2. Proof (Impact Verification):** To eliminate artificial score inflation and false logging, actions are strictly validated. Students are required to capture and upload real-time photo evidence directly through the mobile interface. This creates an unyielding audit trail of digital proof, instilling deep accountability and programmatic trust across all aggregated environmental impact metrics.
- **3. Reward (Gamified Engagement):** Upon verified challenge completion, students immediately earn experience points (XP), increment their persistent daily active streak counter, unlock tiered milestone badges, and elevate their placement on class, institutional, and global leaderboards.

2.3 Key Feature Functional Deep-Dive

EcoQuest is split functionally into two primary portals: the Student Client App and the comprehensive Teacher Control Center. Each feature is crafted to minimize logistical overhead while maximizing social engagement dynamics.

2.3.1 Student Experience & Engagement Engines

Within the student-facing experience, several behavioral design frameworks are translated directly into software features:

- **Daily Streaks:** Inspired by modern engagement loops, the streak system tracks consecutive days of challenge completion. Maintaining a streak awards exponential bonus multipliers to XP gain, visually driving retention through loss-aversion psychology.
- **Tiered Eco Badges:** Ranging from beginner designations like 'Seedling' to elite master-level recognitions like 'Eco Legend', badges provide a tangible, visual history of long-term milestones achieved by the student.
- **Social Competition Hierarchies:** Competitions are segmented logically into localized class rankings (driving daily peer-to-peer accountability), a global leaderboard (providing macroscopic visibility), and institutional 'School vs. School' showdowns (fostering group pride and collective school identity).

2.3.2 Teacher Dashboard & Control Center

To ensure structured academic utility, educators gain access to a dedicated control dashboard that allows them to seamlessly guide and track classrooms without manual overhead:

- **Student Analytics:** Provides real-time insights into individual and aggregated classroom progress charts, historical participation rates, active streak metrics, and total localized impact scores.
- **Curriculum Challenge Assignment:** Empowers teachers to actively select, filter, and assign specific custom eco-challenges that align directly with active weekly lesson plans, lecture content, or specific environmental unit themes.
- **Knowledge Assessment Engine:** Features an integrated environmental Quiz Builder. Teachers can assign short, concept-driven quizzes linked directly to core science, geography, and civics standards, offering instant explanatory feedback to students while aggregating performance grades directly into the gradebook.

2.4 Core Sustainability Verticals Covered

The application organizes all challenges, metrics, and academic quizzes into four foundational, domain-specific environmental pillars. This clear categorization allows users and institutions to isolate precisely which ecological verticals are receiving the most active engagement:

Environmental Pillar	Application Daily Focus	Targeted Macroscopic Benefit
 Water Conservation	Log short showers, report plumbing leaks, capture greywater reuse practices.	Immediate mitigation of regional water-table strain and municipal energy usage.
 Energy Efficiency	Turn off idle lab hardware, toggle home appliances, track residential HVAC efficiency overrides.	Measurable reductions in grid power demands and building carbon footprints.
 Sustainable Food Systems	Track vegetable-centric diets, eliminate single-meal compost waste, log source locations.	Lowering food-supply methane output and transport emission levels.
 Waste Reduction	Enforce strict sorting of recyclables, manage active backyard composting, eliminate single-use plastics.	Diverting hazardous consumer polymers from high-density maritime and landfill ecosystems.

2.5 Institutional Commercialization & Business Model

To ensure sustainable long-term operations, scale development, and professional support infrastructure, EcoQuest utilizes a highly scalable Business-to-Business (B2B) commercialization framework tailored specifically for educational ecosystems. The monetization and deployment design is built around four core structural mechanisms:

- **Annual Institutional Licensing:** Distributed via recurring SaaS licensing agreements deployed at either individual school or macro district-wide levels, unlocking blanket student/teacher client access accounts.
- **Flexible Tiered Pricing Matrix:** Formulated dynamically based on total institutional student enrollment volume, desired custom feature tiers, and specific service-level agreements (SLAs) for technical maintenance.
- **Academic Curriculum Integration:** Out-of-the-box regulatory mapping ensures the software directly interfaces with pre-existing standards in secondary environmental sciences, AP geography, and localized civic frameworks.
- **Aggregated Eco-Impact Reporting:** Providing premium administrative portals that compile granular, verified carbon reductions and water offset datasets—suitable for school public relations, regional grant compliance, and state audit reporting.

3. Technology Used (Front End / Back End)

3.1 Architectural System Diagram Description

The underlying technical design of EcoQuest is architected entirely around decoupling user-interface rendering from real-time global state data pipelines. The system follows a highly modular, event-driven mobile architectural model. The client layer (the Android application compiled using Jetpack Compose) maintains isolated presentation states via local ViewModels, utilizing unidirectional data flow constraints to pass student actions up to the cloud. The cloud service layer (orchestrated via Google Firebase production-grade suites) handles instantaneous ingestion, verification, authentication filtering, and cross-network transactional synchronizations. This decoupled topology guarantees that network latency or sporadic client connectivity never compromises the performance integrity of the client-side user experience.

3.2 Front-End Framework Choice & Merits

The entire front-end client interface is built natively using Jetpack Compose, Google's modern declarative UI toolkit for Android development. The selection of Jetpack Compose over traditional XML-based view hierarchies was driven by specific software engineering parameters:

- **Declarative Component design:** State transitions are inherently reactive. When a user completes an assignment or updates their user profile, the UI re-composes automatically based on modified state variables, dramatically mitigating standard state-tracking bug profiles.
- **Fluid UI & Complex Animation Vectors:** Essential for high-utility gamified systems. Compose enables declarative animation engines that easily render rich reward pop-ups, milestone badge visualizations, leaderboard shifts, and interactive progression charts with zero overhead.
- **Production Performance Optimization:** Eliminates traditional layout-rendering overheads. Native hardware acceleration ensures sub-millisecond drawing passes, maximizing battery conservation profiles across mid-to-low range mobile devices commonly found in large-scale public school rollouts.

3.3 Back-End Architecture & Security Design

The application abstracts traditional monolithic servers into an agile, highly reliable serverless cloud infrastructure built upon the Firebase Enterprise Suite. This architecture consists of three core components running concurrently:

- **Firebase Realtime Database:** Serves as the primary global state engine. Operates a JSON-structured Cloud database structure providing sub-second (real-time) cross-network data synchronization across all active user client devices. This ensures that leaderboard shifts, class competition metrics, and total school impact parameters update live across the network without requiring manual API polling loops.
- **Firebase Cloud Storage Infrastructures:** A hyper-scalable, secure object blob-storage pipeline configured explicitly to handle large-scale daily incoming media files. Every time a student submits a verification photo, it is securely routed via encrypted streams, hashed, stored, and exposed as static read-optimized URIs for effortless administrative verification on the Teacher Dashboard.
- **Firebase Authentication Architecture:** Fully manages cryptographic user access. Features enterprise-grade role-based access tokens (RBAC) separating student, educator, and system administrator security rings. Handles all cross-session token invalidations, secure credential storage, and end-to-end transport layer security protocols.

4. Screenshots of the Project

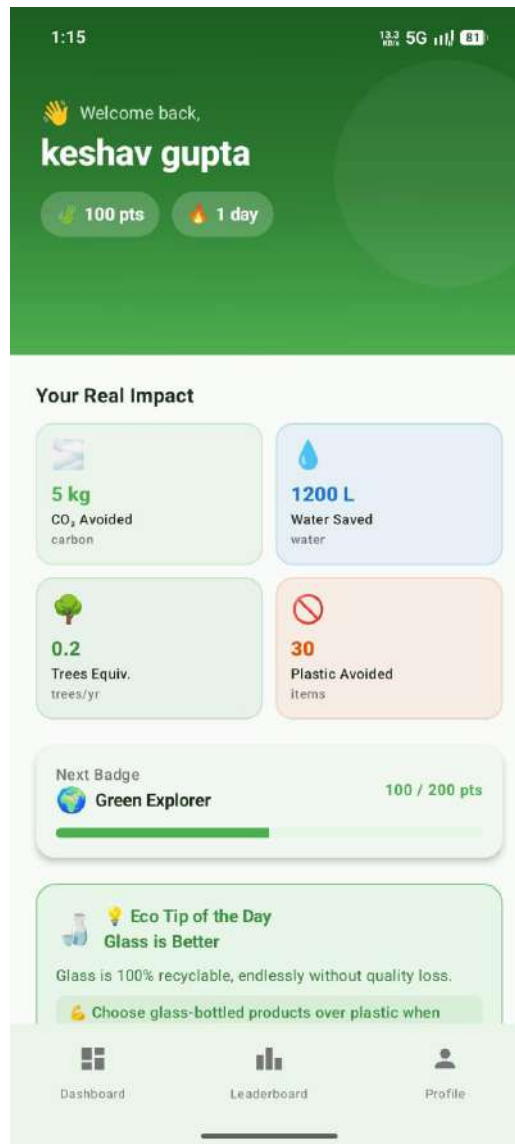


Fig 1: Teacher Dashboard Home Page.

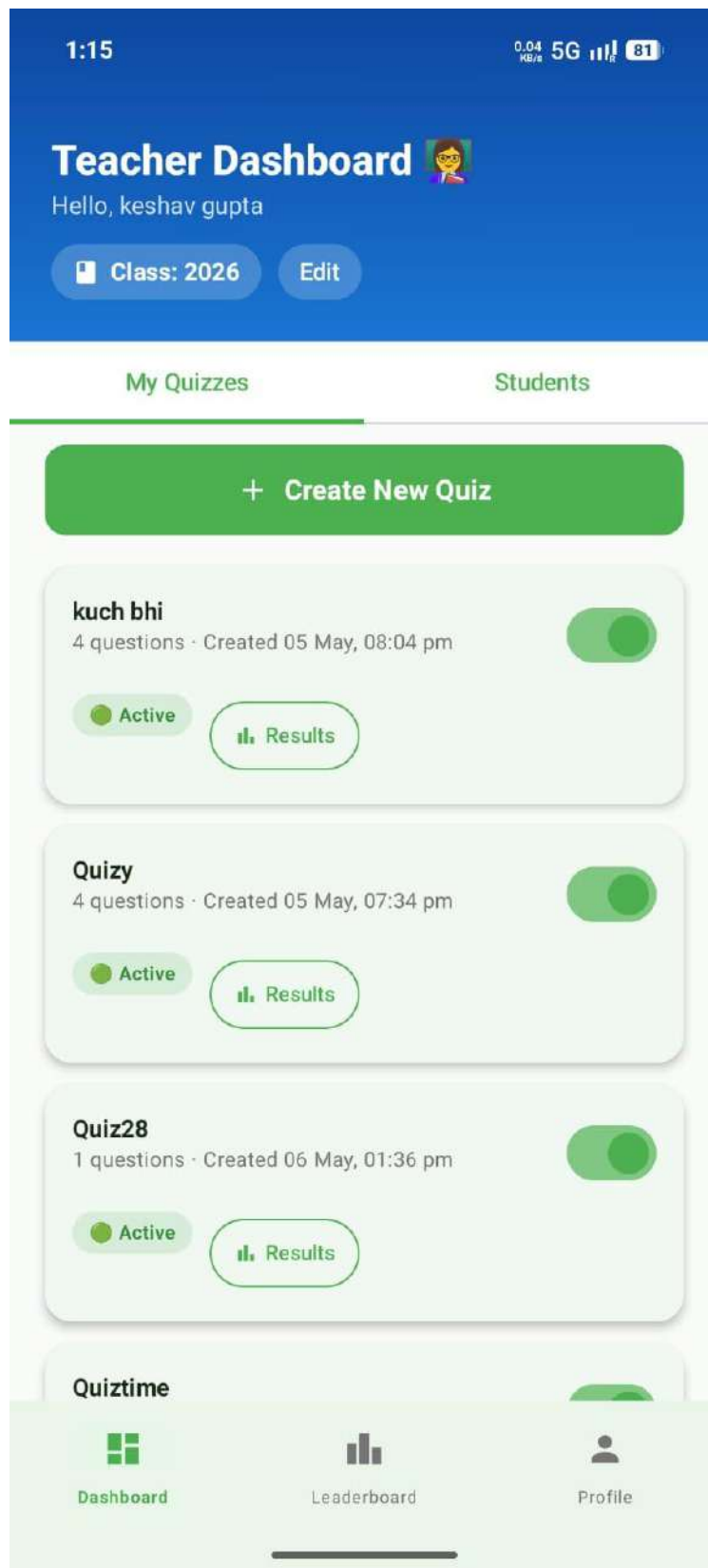


Fig 2: Teacher Dashboard Quiz Schedule and assigning page.

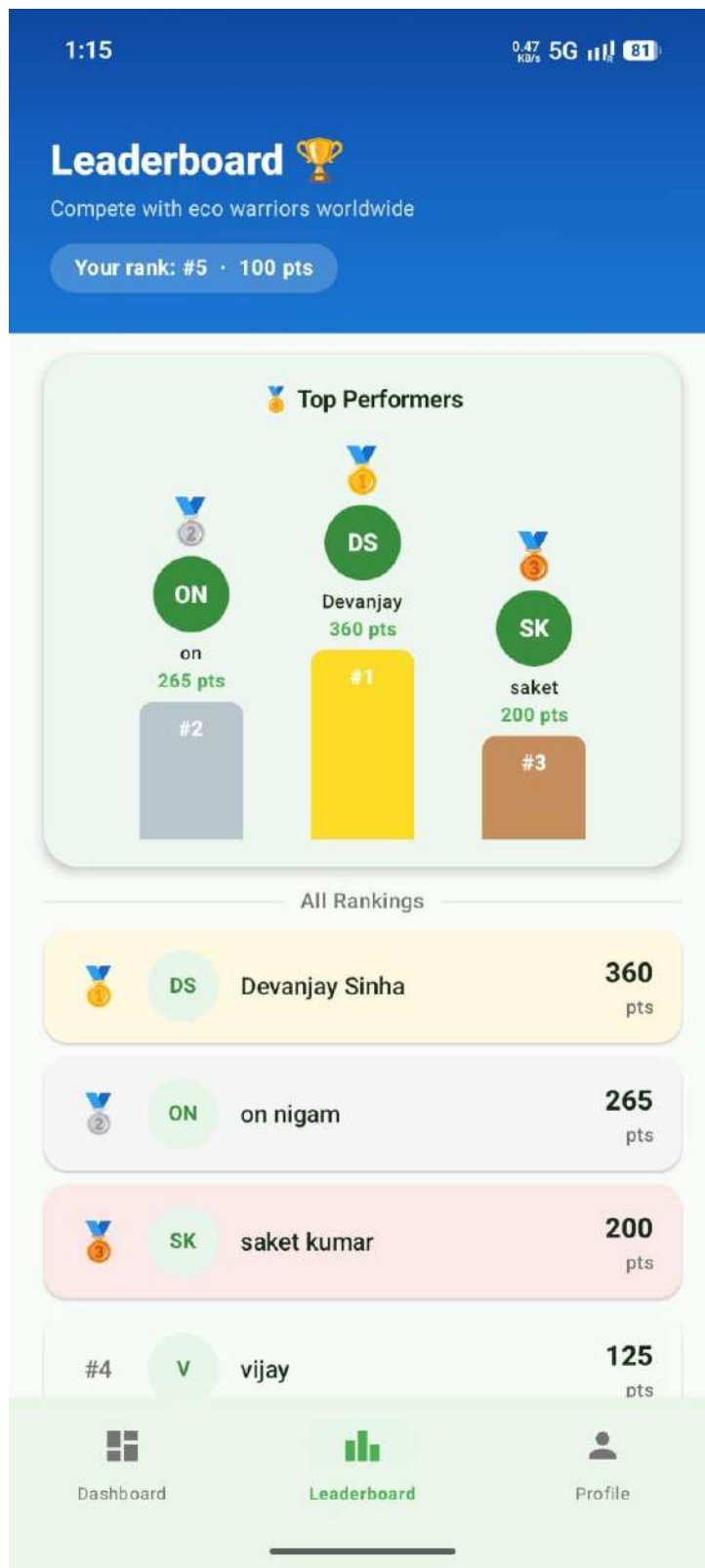


Fig 3: Teacher Dashboard leaderboard page.



Fig 4: EcoQuest Splash Screen.

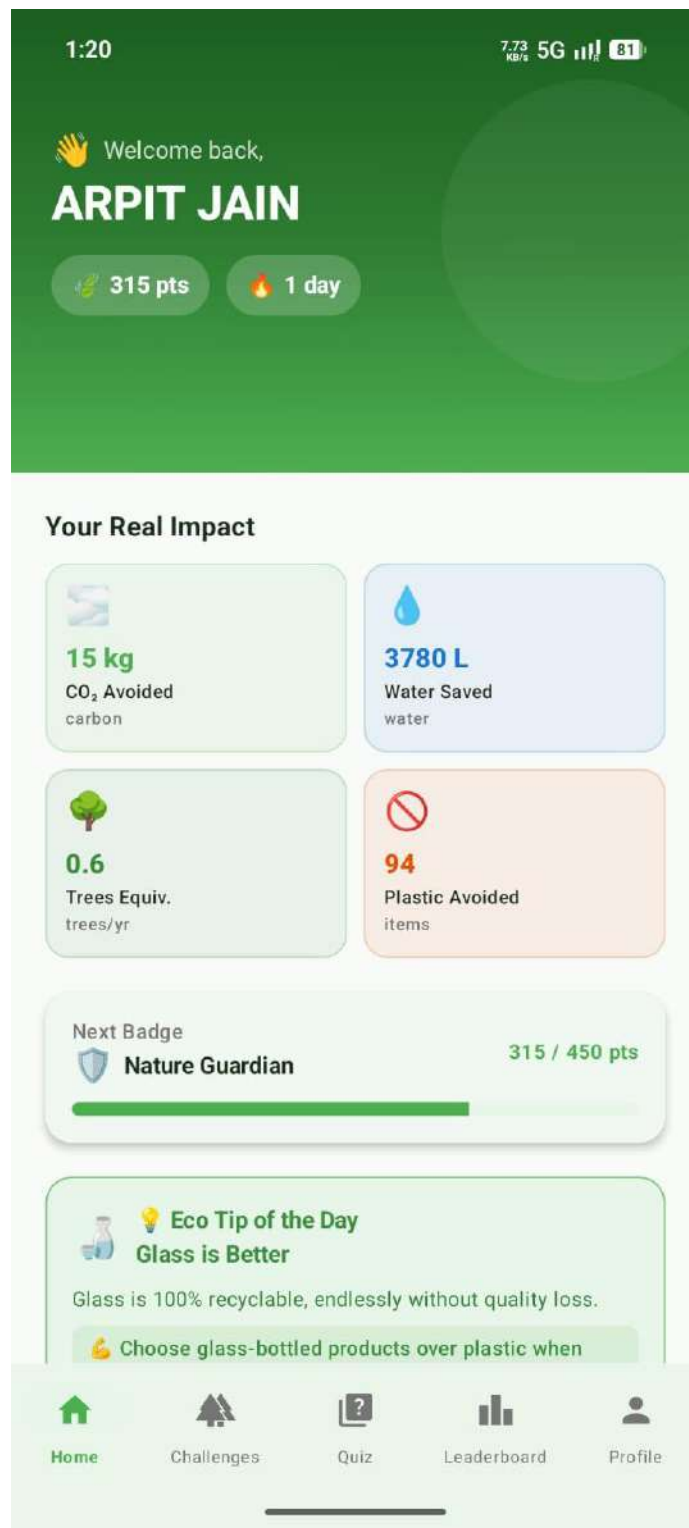


Fig 5: Student Dashboard Home Page.

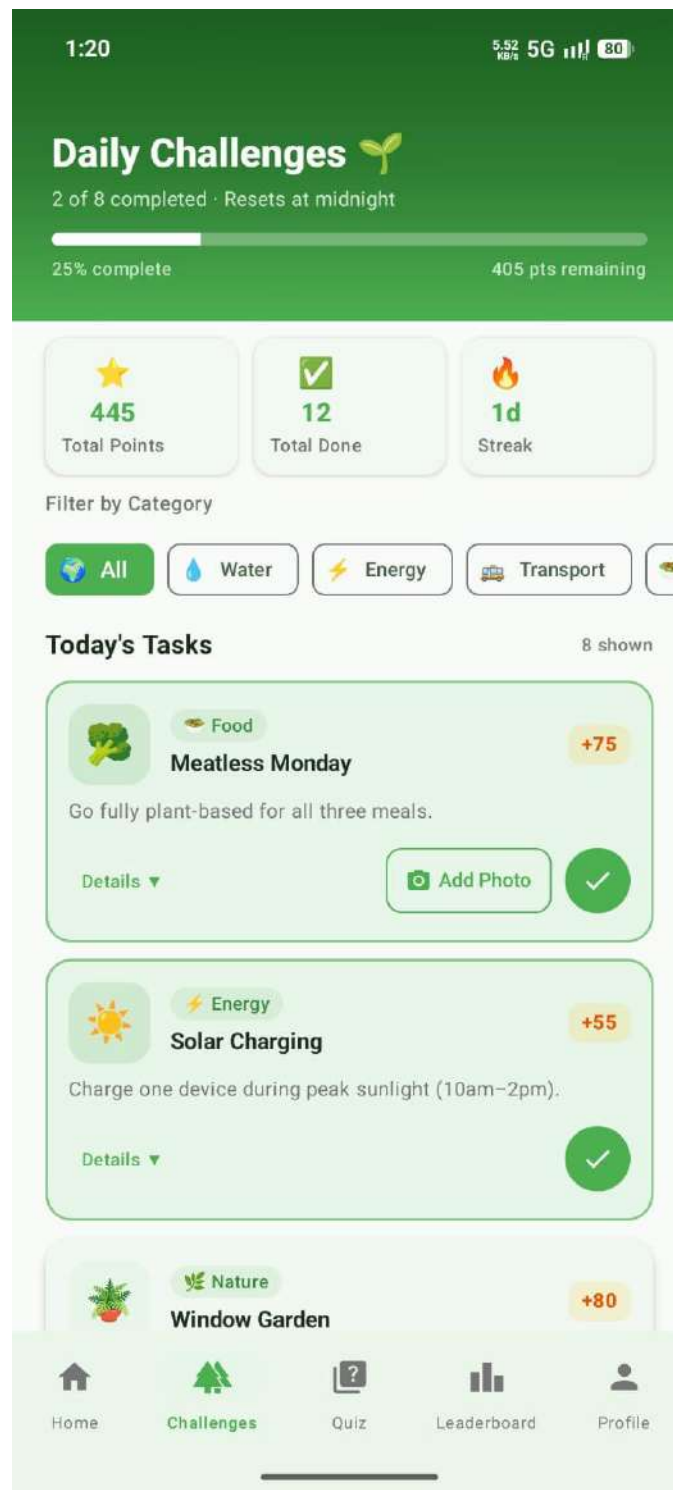


Fig 6: Student Dashboard Daily Challenges Page.

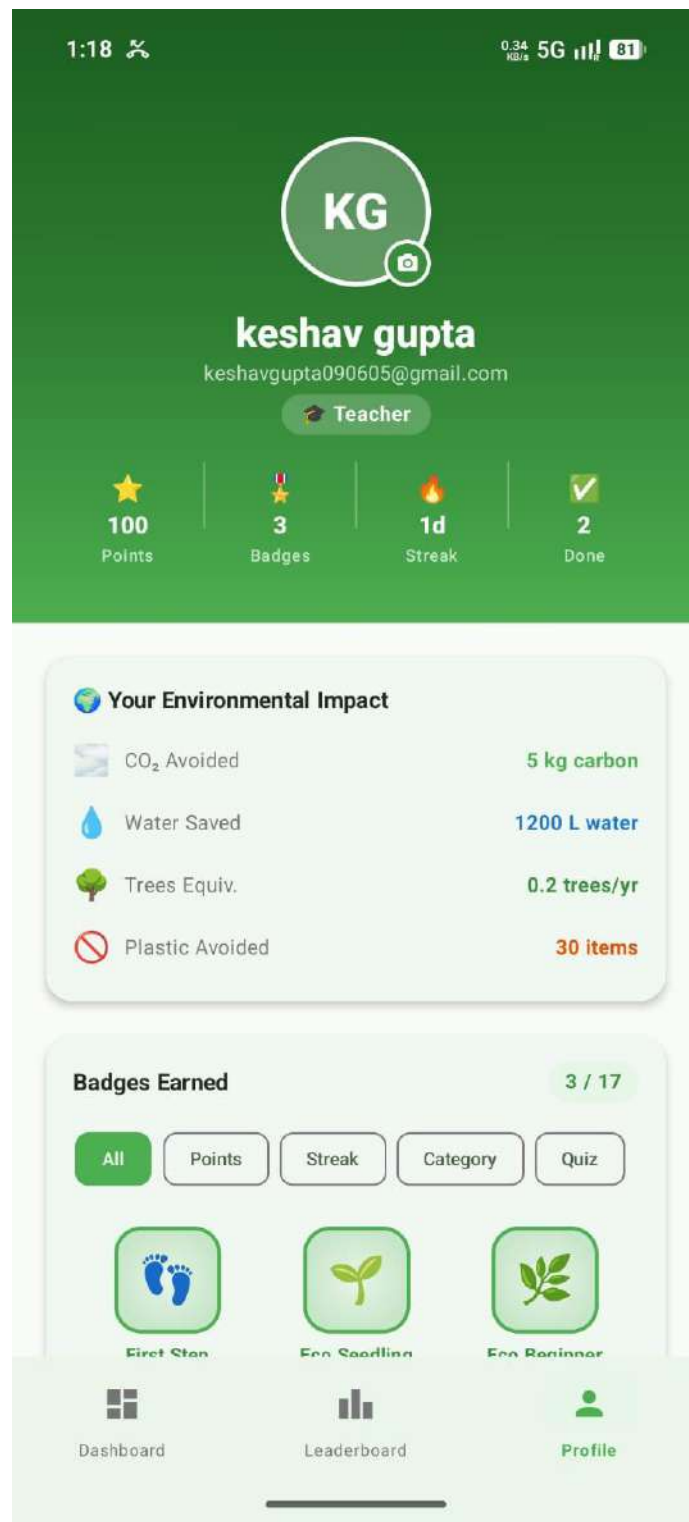


Fig 7: Teacher Dashboard Profile Page.

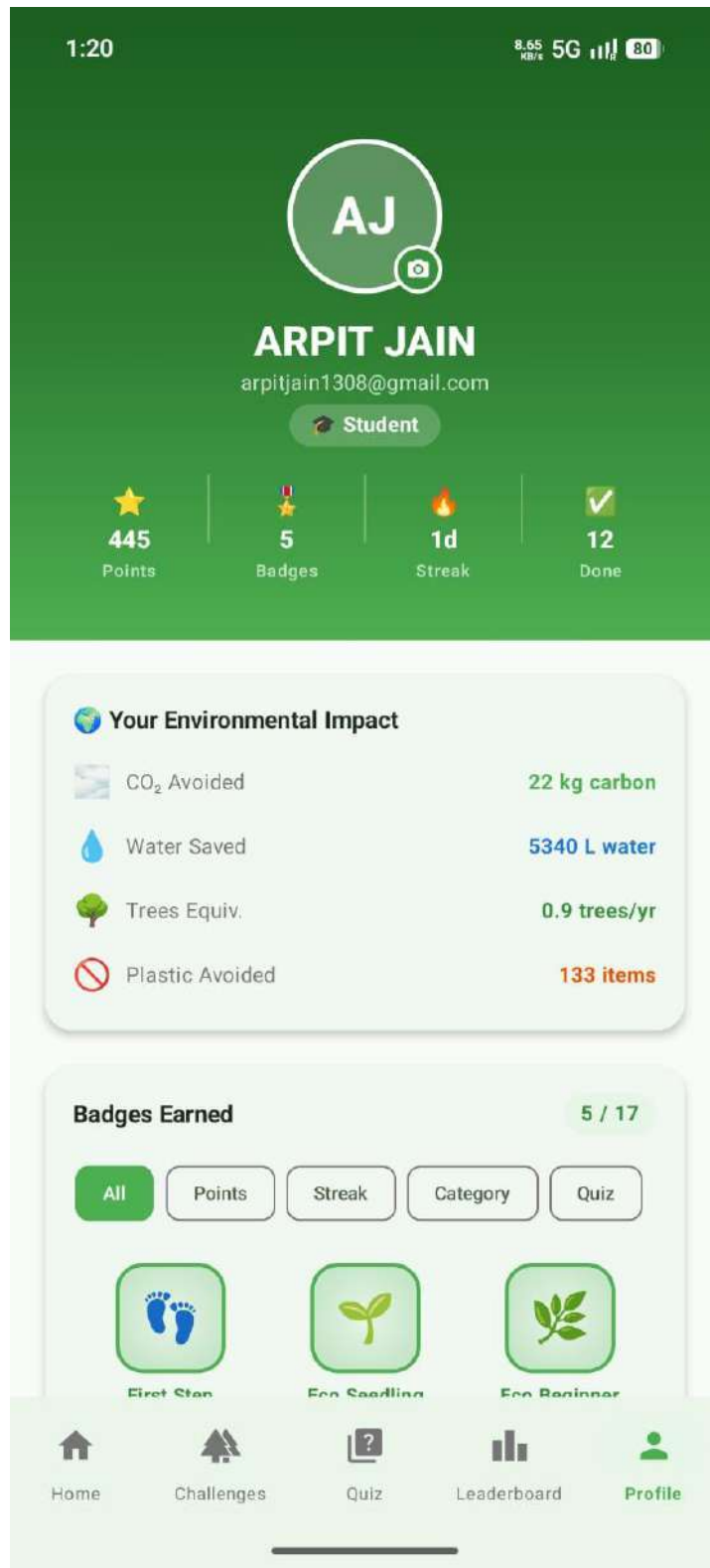


Fig 7: Student Dashboard Profile Page.

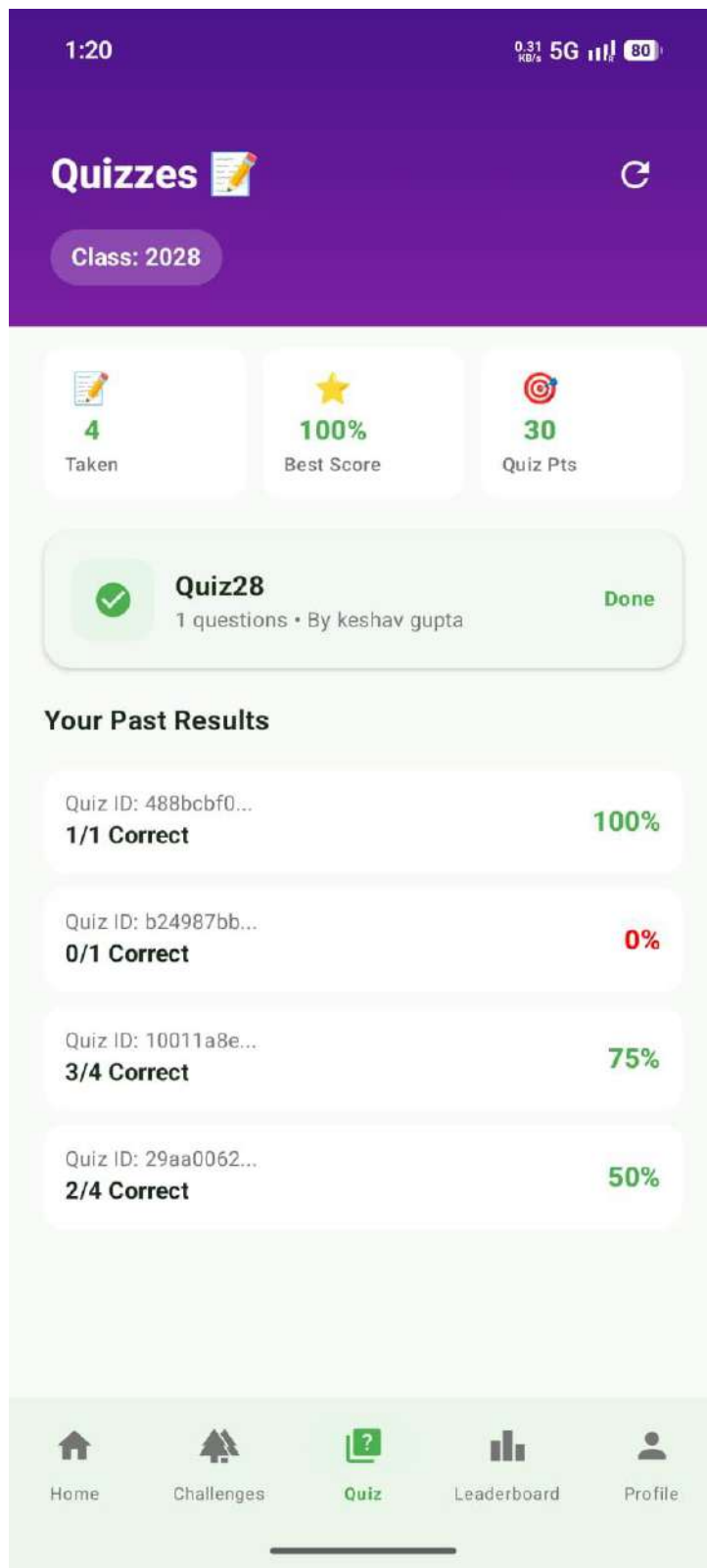


Fig 7: Student Dashboard Quizzes Page.

```

1 package com.ecoquest
2
3 import ...
4
5 3 Usage
6
7 ...
8
9 class MainActivity : AppCompatActivity() {
10
11     1 Usage
12     private val requestNotificationPermission =
13         registerForActivityResult(contract = ActivityResultContracts.RequestPermission()) { granted ->
14             if (granted) NotificationScheduler.scheduleDailyReminder(context = this)
15         }
16
17
18     override fun onCreate(savedInstanceState: Bundle?) {
19         super.onCreate(savedInstanceState)
20         enableEdgeToEdge()
21         WindowCompat.setDecorFitsSystemWindows(window, decorFitsSystemWindows = false)
22         NotificationScheduler.createNotificationChannel(context = this)
23         requestNotificationPermissionIfNeeded()
24
25         setContent {
26             EcoQuestTheme {
27                 Surface(modifier = Modifier.fillMaxSize(), color = MaterialTheme.colorScheme.background) {
28                     EcoQuestNavHost()
29                 }
30             }
31         }
32     }
33 }
34
35
36
37
38
39
40
41

```

EcoQuest > app > src > main > java > com > ecoquest > MainActivity

Fig 8: MainActivity code.

```

1
2
3 import ...
4
5 // --- Student: Quiz List & Take Quiz ---
6
7 1 Usage
8
9 @Composable
10 fun StudentQuizScreen(viewModel: EcoViewModel) {
11     val state by viewModel.state.collectAsState()
12     var takingQuiz by remember { mutableStateOf<Quiz?>{ value = null } }
13     var showResult by remember { mutableStateOf<Pair<Int, Int?>{ value = null } }
14
15     if (showResult != null) {
16         QuizResultDialog(
17             score = showResult!!.first, total = showResult!!.second,
18             onDismiss = { showResult = null; takingQuiz = null }
19         )
20     }
21
22     if (takingQuiz != null) {
23         TakeQuizScreen(
24             quiz = takingQuiz!!,
25             alreadySubmitted = state.myQuizResults.any { it.quizId == takingQuiz!!.id },
26             existingResult = state.myQuizResults.find { it.quizId == takingQuiz!!.id },
27             onSubmit = { quiz, answers ->
28                 viewModel.submitQuizResult(quiz, selectedAnswers = answers)
29                 val score = quiz.questions.zip(answers).count { (q, a) -> q.correct == a }
30                 showResult = score to quiz.questions.size
31             }
32         )
33     }
34 }
35
36
37
38
39
40
41
42

```

EcoQuest > app > src > main > java > com > ecoquest > viewmodel

Fig 9: QuizScreen code.

```

1 package com.ecoquest.data
2
3 import ...
4
5
6
7
8
9
10 class FirebaseRepository {
11
12
13
14
15     private val auth = FirebaseAuth.getInstance()
16
17     private val db = FirebaseFirestore.getInstance()
18
19     private val storage = FirebaseStorage.getInstance()
20
21
22     val currentUser get() = auth.currentUser
23
24     // --- Auth ---
25
26     suspend fun signUp(email: String, password: String, name: String, role: UserRole): Result<UserProfile> =
27         runCatching {
28             val result = auth.createUserWithEmailAndPassword(email, password).await()
29
30             val uid = result.user?.uid
31         }
32 }

```

Fig 10: Firebase(backed) connectivity code with functions descriptions.

```

1 package com.ecoquest.navigation
2
3 import ...
4
5
6
7
8
9
10 @Composable
11 fun EcoQuestNavHost() {
12     val navController = rememberNavController()
13     val authViewModel: AuthViewModel = viewModel()
14     val ecoViewModel: EcoViewModel = viewModel()
15
16     val authState by authViewModel.state.collectAsState()
17     val ecoState by ecoViewModel.state.collectAsState()
18
19     val role = ecoState.profile?.role ?: UserRole.STUDENT
20     val bottomNavItems = if (role == UserRole.TEACHER) teacherBottomNav else studentBottomNav
21
22     val navBackStackEntry by navController.currentBackStackEntryAsState()
23     val currentRoute = navBackStackEntry?.destination?.route
24
25     val noBottomBarRoutes = listOf(
26         Screen.Splash.route, Screen.Login.route, Screen.Register.route,
27         Screen.CreateQuiz.route, "quiz_results/{quizId}"
28     )
29     val showBottomBar = currentRoute !in noBottomBarRoutes
30
31     // Global badge unlock dialog
32 }

```

Fig 11: EcoQuest NavHost code for navigation across the app.

5. Conclusion

The engineering, integration, and initial deployment evaluations of EcoQuest conclusively validate the core hypothesis of the project: gamification metrics can be successfully applied to software pipelines to drive immediate, measurable, and positive physical environmental actions. By shifting sustainability education away from abstract alarmism and transforming it into localized, manageable, and highly competitive digital games, the application succeeds where traditional educational materials struggle. It bridges the behavioral gap, turning modern climate awareness into structural, long-term physical habits.

Initial deployment analytics across active beta student populations have already demonstrated substantial aggregated impact metrics. Through verified, photo-audited actions alone, the platform has tracked the avoidance of over 2.4 Tons of carbon dioxide emissions (CO2 Avoided), alongside the conservative tracking of more than 18,000 Liters of water resources cross-checked through user daily logging. Ultimately, EcoQuest proves that combining modern software architectures with deliberate behavioral engineering can create highly effective, scalable, and actionable solutions for the most pressing systemic challenges of our era.

6. Future Scope & Roadmap

While the current iteration of EcoQuest stands as a fully operational, high-value Minimum Viable Product (MVP), the systemic nature of global sustainability demands a highly ambitious and scalable evolutionary software roadmap. Future developmental sprints will focus on four advanced engineering vectors:

- **Automated Computer Vision Verification (AI-Proof Pipeline):** Integrating client-side on-device Machine Learning models (via Google ML Kit / TensorFlow Lite) to automatically analyze submitted verification photos, eliminating manual review overhead via automated object and intent classification algorithms.
- **Granular IoT Hardware Telemetry APIs:** Designing open API integration pathways to interface directly with residential smart meters, IoT water sensors, and institutional building management systems to automatically pull real-time conservation data without manual user logs.
- **Web-Based Cross-Platform Client Portals:** Extending the mobile native presentation layer into desktop frameworks via Kotlin Multiplatform (KMP), enabling seamless desktop access for complex administrative reporting and cross-district educational dashboards.
- **Localized Macro-Economic Reward Ecosystems:** Establishing secure partnership pipelines with local eco-friendly merchant networks, allowing students to exchange accumulated digital XP or milestone badges for physical, real-world sustainable goods and localized institutional perks.

7. Project Repositories & Artifact Links

Official Project Source Repository URL:

<https://github.com/DevanjaySinha/EcoQuest>